Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **React**, **JavaScript**, and TailwindCSS web applications, healthcare platforms, and **search**-driven content tools, with production support, **Agile** delivery, and secure **SDLC** practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a multi-tenant education platform on Azure SQL and Blob Storage with **row-level security** and normalized schemas for 500+ users, tuning queries and indexing to cut dashboard latency from 1.2s to 350ms.
- Maintained production data integrations with validation, reconciliation checks, error handling, and CI/CD **release workflows**, processing 5K+ monthly jobs at 99%+ reliability across PostgreSQL, SQL Server, and cloud storage.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Built a **React** dashboard for file upload, grading history, and question management backed by modular Flask APIs, serving 150+ instructors and 10K+ daily requests while cutting feedback turnaround 60%.
- Supported deployment and **production reliability** for instructor-facing grading, securing PDF, DOCX, and ZIP ingestion and adding OCR fallback paths that kept API responses stable during peak load.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Built object-oriented API modules with Singleton and Factory **design patterns** and AWS Glue transformations, improving response times 30% and pipeline efficiency 68% while cutting technical debt 20%.
- Managed feature, development, staging, and production branches in GitHub with **Jenkins CI/CD** and nightly unit and integration tests, collaborating with Microsoft and GitHub stakeholders through Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed **React** and Node.js features for a MERN resource platform with WebSocket updates, JWT authentication, and **role-based access**, synchronizing 2M+ records across locations for core business workflows.
- Monitored production with Sentry, Azure Monitor, and Kibana and used **Google Analytics** load-time and drop-off insights to guide fixes, reducing resource wastage 42% and increasing lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client **requirements** into an end-to-end architecture with Figma designs and 12+ mapped integration points, presenting the solution to stakeholders and reducing design errors 25% before development.
- Built Flask APIs and Firebase real-time sync behind React and **TypeScript** web clients, sustaining sub-200ms booking responses across 50K+ records and improving cross-platform compatibility 15% across mobile and web.

Skills

Web Development: React.js, JavaScript, TypeScript, HTML, CSS, TailwindCSS, jQuery, Next.js, Node.js, Express.js, REST APIs
Search & Data: Semantic Search, Metadata Indexing, FAISS, Pinecone, SQL, PostgreSQL, SQL Server, MySQL, MongoDB
Practices: SDLC, Design Patterns, OOP, Git, GitHub, Agile/Scrum, CI/CD, Unit Testing, Code Reviews, RBAC, Python
Support & Cloud: Production Support, Troubleshooting, Sentry, Azure Monitor, Google Analytics, Jenkins, Docker, Azure, AWS

Projects

HMIS - Hospital Management Information System Live | *TypeScript, MERN, TailwindCSS, ABDM APIs, Twilio, Sentry*

- Built a responsive **TailwindCSS** and TypeScript **healthcare** platform with consent-based patient records, appointment dashboards, Twilio reminders, and Sentry monitoring, reducing administrative workload 30% across 100+ daily appointments.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, Convex, Shadcn*

- Built a secure document management and **semantic search** platform with Clerk authentication, FAISS and Pinecone retrieval, and a Shadcn interface, reaching 95% query accuracy across 50K+ documents for 200+ users.

PiXMod Live | *Next.js, MongoDB, CLIP, SAM, Stable Diffusion, Stripe*

- Built a Next.js image platform with MongoDB asset and metadata storage and CLIP content-based **image search** across 100K+ indexed embeddings, plus Stripe subscriptions that raised premium conversion 18%.